



MARTIN-LUTHER-UNIVERSITÄT
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ADVANCED MICROECONOMICS

Decisions under Risk and Uncertainty I - Expected Utility

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Outline

1) Consumer Theory

- 1.1) Preferences and Utility (L1)
- 1.2) Utility Maximization (L2)
- 1.3) Expenditure Minimization (L3)
- 1.4) Demand Functions and Duality (L4+L5)

2) General Equilibrium

- 2.1) Exchange Economy (L6)
- 2.2) Welfare Economics and General Equilibrium (L7)

3) Decisions under Risk and Uncertainty

- 3.1) Expected Utility (L8)
- 3.2) Risk Preferences (L9)
- 3.3) Insurances (L10)
- 3.4) Safety Regulations and the Value of a Statistical Life (11)

4) Behavioral Economics (L12)

Literature

- Autor, David (2016): Lecture Notes 16
- Anger, Erik (2021): A Course in Behavioral Economics, 3rd edition, Palgrave MacMillan, Chapter 4 and 6

Decision under risk and uncertainty

- **So far:** Models of decisions in situations **without uncertainty**, more specifically, decisions about alternatives that had certain outcomes at time of decision making
 - In **reality**, many decisions are taken under risk and uncertainty because they are taken for the future
 - **Events** such as policy changes, accidents, new viruses happen, due to which we cannot be certain at the time of the decision what the outcomes of the different alternatives are going to be
- The actual utility of the different alternatives depends on the **state of the world** that is realized

Examples for decision under risk and uncertainty

- Are you going to bike or take public transport to university?
- Are you going to pursue a Ph.D.? In which subject? At which university?
- Are you going to marry this person? Or that? Or not at all?
- Are you going to take up a supplemental health insurance? A disability insurance? A legal expenses insurance?
- Are you buying a new car or a used one (or non at all)?
- Are you participating in an in-person lecture during a pandemic?

Risk or uncertainty

The typical distinction (following Knight, 1921)

Decision under uncertainty

Probabilities of different possible states of the world are unknown.

Example: At the beginning of the COVID-19 pandemic completely unclear what could happen with which probability (e.g. school closures)

Decision under risk

Different possible states of the world and the probability with which they will occur are known (to the decider).

Example: Decision to get a flu shot / COVID-vaccine.

- Extremes are boundary cases; in reality we typically observe mixtures, e.g.
 - probabilities with imprecision intervals
 - subjective probabilities

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Decisions under Uncertainty

- **Uncertainty:** Probabilities are not known (or not defined), but the outcomes for the different possible states of the world are known.

Decision Rules

- 1 **Maximin:** choose the alternative with the **largest minimum** payoff
- 2 **Maximax:** choose the alternative with the **largest maximum** payoff (a.k.a. the “YOLO criterion”)
- 3 **Minimax Regret:** choose the alternative for which the maximum possible regret is minimal

Decisions under Uncertainty

- **Example:** Should I take an umbrella with me?

- Alternatives and states of the world

	Rain	No Rain
With umbrella	dry, unhappy	dry, unhappy
Without umbrella	wet, miserable	dry, happy

- Corresponding payoffs

	Rain	No Rain
With umbrella	3	3
Without umbrella	0	5

Decisions under Uncertainty — Solution (Umbrella)

	Rain	No Rain
With umbrella	3	3
Without umbrella	0	5

Maximin (*choose the best worst-case outcome*):

$$\min(\text{With}) = 3, \quad \min(\text{Without}) = 0 \Rightarrow \textbf{With umbrella.}$$

Maximax (*choose the best best-case outcome*):

$$\max(\text{With}) = 3, \quad \max(\text{Without}) = 5 \Rightarrow \textbf{Without umbrella.}$$

Minimax Regret (*minimize the maximum possible regret*):

	Rain	No Rain
Best payoff in each state	3	5
Regret With umbrella	$3 - 3 = 0$	$5 - 3 = 2$
Regret Without umbrella	$3 - 0 = 3$	$5 - 5 = 0$

$$\max \text{Regret}(\text{With}) = 2, \quad \max \text{Regret}(\text{Without}) = 3 \Rightarrow \textbf{With umbrella.}$$

Decisions under Uncertainty

- Maximin is probably the most widely used criterion.
- But it is not always reasonable, since it only considers the smallest possible payoff in each state of the world — no matter how small the probability of that state actually is.

Harsanyi's Challenge

A New Yorker receives two job offers:

- New York: guaranteed income of €50,000
- Chicago: income of €100,000, but an extremely small risk of a fatal plane crash on business trips

Maximin would always recommend choosing New York (no matter how small the crash probability is).

→ When probabilities are known, it seems sensible to take them into account when making decisions!

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Recap: Probability Judgement

- Decisions under risk are based on **probability theory**.
 - Normative theory of assessing probabilities (*“probabilistic judgment”*)
- **Basic concepts:**
 - **Probability:** a function $Pr(A)$ that assigns a number between 0 and 1 to each event
 - *Unconditional probability:* the probability of an event, independent of other events
 - *Conditional probability:* the probability of A given that B has occurred, $Pr(A|B)$
- Like rational choice theory, it is **axiomatic** – based on a few plausible fundamental assumptions
- For review: Angner (2021), Chapter 4

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Simple Lotteries

- Assumption: We have a finite number of states of the world (outcomes): $N < \infty$ with known probability for each outcome n : $p_n \geq 0$

Simple Lottery

List of probabilities, $L = (p_1, \dots, p_N)$, where $p_n \geq 0$ for all n and $\sum_n p_n = 1$

- **Decision situation:** Choice between multiple lotteries
- **The outcomes of a simple lottery are certain.**

Compound Lotteries

- The outcomes of a lottery do not have to be certain.
- Instead, they may itself be a lottery.
- If that is the case, the lottery is called a **compound lottery**.

Compound Lottery

Given K simple lotteries $L_k = (p_1^k, \dots, p_N^k)$ for $k = 1, \dots, K$ and probabilities $\alpha_k \geq 0$ with $\sum_{k=1}^K \alpha_k = 1$, the **compound lottery** $(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$ is the risky alternative that yields the simple lottery L_k with probability α_k for $k = 1, \dots, K$.

Lotteries – simple and compound

Examples

Decision between two simple lotteries A and B:

Should you study on the day before the exam or go to your friends dinner party?

- A** Study: with 95% probability good grade, with 5% failed exam.
- B** Dinner party: with 30% probability good grade, with 70% failed exam.

Compound lottery for alternative A:

- A** Study:
 - with 10% probability you'll study with a headache, therefore studying is not effective and probability of good grade is 50%, probability of failed exam 50%.
 - with 90% no headache, then 100% probability of good grade.

Assumption of consequentialism

- Every compound lottery can be reduced to a simple lottery, which is then called the **reduced lottery**.

Reduced Lottery

In the reduced lottery only the probabilities of the final (certain) outcomes (good grade/fail in example) are shown.

- We can show that decision making is only influenced by probabilities of the final (certain) outcomes:

Assumption of consequentialism

Only reduced lotteries matter for individual decision making

- With this assumption it suffices to think about preferences over reduced lotteries.

Reduced Lottery

Example

Compound and reduced lottery for alternative A:

- Compound lottery (Study):
 - with 10% probability **headache**, therefore studying not effective and probability of good grade 50%, probability of failed exam 50%:
 - probability of good grade: $0.1 \times 0.5 = 0.05$
 - probability of failing: $0.1 \times 0.5 = 0.05$
 - with 90% **no headache**, then 100% probability of good grade:
 - probability of good grade: $0.9 \times 1 = 0.9$
 - probability of failing: $0.9 \times 0 = 0$
- Reduced lottery (Study):
 - probability of good grade: 0.05 (studying with headache) + 0.9 (studying without headache) = 95%.
 - probability of failing: 0.05 (due to headache when studying) + 0 = 5%.

Example

Look at the following simple lotteries

$$L_1 = (1, 0, 0),$$

$$L_2 = (1/4, 3/8, 3/8),$$

$$L_3 = (1/2, 1/2, 0),$$

$$L_4 = (1/2, 0, 1/2)$$

with the compound lotteries

$$L_{12} = (L_1, L_2; 1/3, 2/3),$$

$$L_{34} = (L_3, L_4; 1/2, 1/2).$$

What are the reduced lotteries of L_{12} and L_{34} ? Interpret your results!

Example – Solution

Reduced lottery of L_{12} is

$$\tilde{L}_{12} = \left(\frac{1}{3} \cdot 1 + \frac{2}{3} \cdot \frac{1}{4}, \frac{1}{3} \cdot 0 + \frac{2}{3} \cdot \frac{3}{8}, \frac{1}{3} \cdot 0 + \frac{2}{3} \cdot \frac{3}{8} \right).$$

$$\tilde{L}_{12} = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4} \right).$$

Reduced lottery of L_{34} is

$$\tilde{L}_{34} = \left(\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2}, \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 0, \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{1}{2} \right).$$

$$\tilde{L}_{34} = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4} \right).$$

Interpretation:

Both lotteries L_{12} and L_{34} reduce to the *same* reduced lottery.

⇒ Rational decision maker should be **indifferent** between L_{12} and L_{34} .

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Expected Value

The expected value $EV(L)$ of a lottery L is the average value that can be expected from this lottery.

- Suppose we have a lottery L with possible events $S_1, \dots, S_n$, probabilities $Pr(S_1), \dots, Pr(S_n)$, and outcomes $C_1, \dots, C_n$.
- The expected value of L is:

$$\begin{aligned} EV(L) &= Pr(S_1) \times C_1 + Pr(S_2) \times C_2 + \dots + Pr(S_n) \times C_n \\ &= \sum_{j=1}^n Pr(S_j) \times C_j \end{aligned}$$

Expected value rule

Decision between lotteries is made by maximizing expected value of the lotteries (μ -rule).

Example

- Choice of **beverage for selling at student party**: beer or mull wine - payoff (total sales) depends on weather

S	Sun	Rain	Snow
P(S)	0.4	0.3	0.3
a1: Beer	200	120	20
a2: Mull wine	20	100	220



- Expected payoffs

$$EV(a1) = 0.4 \times 200 + 0.3 \times 120 + 0.3 \times 20 = 122$$

$$EV(a2) = 0.4 \times 20 + 0.3 \times 100 + 0.3 \times 220 = 104$$

→ You should go for **beer**.

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Information and Learning

- Decisions under uncertainty rely on **imperfect, incomplete, or subjective beliefs** about states of the world.
- Sometimes additional information about the realization of states (i.e. reduction of risk by updated/more reliable probabilities) can be (costly) acquired
- Examples: weather forecasts, medical test results, market analyses

Bayesian Learning / Updating

Bayesian learning is process of updating subjective probabilities of states S based on evidence (E) according to **Bayes' Rule**:

$$Pr(S | E) = \frac{Pr(E | S) Pr(S)}{Pr(E)}$$

Example – continued

- **Bayesian learning** about the probability of **sun** based on a weather app forecasting sun (F^+).
- Prior probabilities:

$$Pr(Sun) = 0.4, \quad Pr(Rain) = 0.3, \quad Pr(Snow) = 0.3$$

- Forecast quality (likelihoods):

$$Pr(F^+ | Sun) = 0.9, \quad Pr(F^+ | Rain) = 0.2, \quad Pr(F^+ | Snow) = 0.1$$

- Total probability that the app forecasts sun:

$$Pr(F^+) = 0.9 \cdot 0.4 + 0.2 \cdot 0.3 + 0.1 \cdot 0.3 = 0.45$$

Example – Bayesian updating

- Updated probability of sun given that the app forecasts sun:

$$Pr(Sun | F^+) = \frac{Pr(F^+ | Sun)Pr(Sun)}{Pr(F^+)} = \frac{0.9 \cdot 0.4}{0.45} = 0.8$$

- Similarly:

$$Pr(Rain | F^+) = \frac{0.2 \cdot 0.3}{0.45} \approx 0.13, \quad Pr(Snow | F^+) = \frac{0.1 \cdot 0.3}{0.45} \approx 0.07$$

- Updated expected payoffs** given forecast “sun”

$$EV(a_1 = \text{Beer} | F^+) = 0.8 \cdot 200 + 0.1333 \cdot 120 + 0.0667 \cdot 20 \approx 177.3$$

$$EV(a_2 = \text{Mull wine} | F^+) = 0.8 \cdot 20 + 0.1333 \cdot 100 + 0.0667 \cdot 220 = 44$$

Role of Baseline Probabilities: Rare Disease Example

- Bayesian updating is strongly driven by the **prior probability** of a state.
- Even highly reliable tests can have **low predictive value** when the condition is rare.
- Consider a disease with prevalence $Pr(D) = 0.1\% = 0.001$
- Medical test with $Pr(\text{Positive} \mid D) = 0.99$ (1% false positive rate)
- Total probability of a positive test:

$$Pr(Pos) = 0.99 \cdot 0.001 + 0.01 \cdot 0.999 = 0.01098$$

- Posterior probability of actually having the disease after a positive test:

$$Pr(D \mid Pos) = \frac{Pr(Pos \mid D) Pr(D)}{Pr(Pos)} = \frac{0.99 \cdot 0.001}{0.01098} \approx 0.090 \quad (= 9\%)$$

Key insight

Even with a highly accurate test, a positive result is very unlikely to indicate disease when the **baseline probability is extremely low**. This is the central challenge in screening for rare conditions.

Value of Information

- The **Expected value of Information (EVI)** measures how much better, on average, a decision-maker performs when additional information E is available.
- Formally, EVI is the **maximum willingness to pay** for access to a signal E before choosing an action a
- Information has **weakly positive value**, i.e. it may be **irrelevant**, but cannot make the decision-maker worse off.
- Intuition: The more information changes the chosen action, the higher its **informational value**.
- Information can differ in its **informativeness**:
 - **Perfect information** reveals the state S with certainty

$$Pr(E|S) = 1$$
 - **Imperfect information** provides only a noisy signal

$$Pr(E|S) \in (0, 1)$$
- Perfect information provides an upper bound for expected value of information.

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The St. Petersburg Game

- St. Petersburg Game

We toss a fair coin until it shows heads. You receive €2 if heads appears on the first toss; if the first toss is tails, you get €4 for heads on the second toss; for every additional toss the payout doubles.

	T	HT	HHT	...
Payout	€2	€4	€8	...
$Pr()$	$\frac{1}{2}$	$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$	$\frac{1}{8}$	...

- What is the expected value of this game?

$$EV = \frac{1}{2} \times 2 + \frac{1}{4} \times 4 + \frac{1}{8} \times 8 + \dots = 1 + 1 + 1 + 1 + \dots = \infty$$

- Would you play this game with me for an entry fee of €1000?*
- What would the **expected value rule**, i.e. choose the option with the highest expected value, predict?

St. Petersburg Paradox

St. Petersburg Paradox

Describes the failure of the **expected value rule** in the **St. Petersburg Game**.

- The paradox goes back to **Nikolaus I Bernoulli**.
 - A game with infinite expected value (like the coin game) should rationally be worth an infinite amount.
 - Problem: The expected value alone does not describe how people perceive money and risk.
- **Daniel Bernoulli** (cousin of Nikolaus) introduced the idea of a **utility function** (instead of monetary value) to measure *happiness*.

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Expected Utility

- **Rational behavior:** Individuals have **preferences** over lotteries, but building preference relations over all possible lotteries is impractical.
- This brings us back to the **concept of (cardinal) utility** only that we define utility function $u()$ over lotteries instead of over bundles of goods.
- Basic idea about expected utility maximization (*Individuals choose the alternative that maximizes expected utility*) goes back to **Daniel Bernoulli**.
- Extension of the basic idea into a **expected utility theory** by **von Neumann and Morgenstern (vNM)** in 1940s allows modeling decision under risk based on utility functions over the certain outcomes of reduced lotteries.

vNM Expected utility form

Von Neumann Morgenstern (vNM) Expected Utility

The utility function $U : \mathcal{L} \rightarrow \mathbb{R}$ has the **vNM expected utility form** if there is an assignment of numbers $u_1, \dots, u_N$ to the N outcomes such that for every simple lottery $L = (p_1, \dots, p_N) \in \mathcal{L}$ we have

$$U(L) = \sum_j^N p_j u_j = u_1 p_1 + \dots + u_N p_N.$$

- The function U is called **von Neumann-Morgenstern (vNM) utility function** or simply **expected utility function**. To distinguish this from the standard consumer theory, it's sometimes also denoted EU .
- vNM representation only works if preferences of rational decision makers have additional properties (i.e., fulfill additional axioms)

Expected utility theorem - Formal

- The decision maker (DM) chooses among a set of **simple lotteries** $\mathcal{L}$.
- **Standard Rationality Assumptions:** The DM has a **complete** and **transitive** preference relation $\succsim$ over $\mathcal{L}$.
- **Additional vNM Axioms:**

- 1 **Continuity:** If $A \succ B \succ C$, then there exists a probability $p \in (0, 1)$ such that

$$B \sim pA + (1 - p)C.$$

- 2 **Independence:** For all lotteries A, B, C and all $\alpha \in (0, 1)$,

$$A \succsim B \iff \alpha A + (1 - \alpha)C \succsim \alpha B + (1 - \alpha)C.$$

$\Rightarrow$ Under these axioms, the preference relation has a **von Neumann–Morgenstern (vNM) expected utility representation**

Continuity

- **Continuity intuition:** Small changes in probabilities do not lead to sudden reversals in preference rankings.
- **Formal definition (vNM):** If $A \succ B \succ C$, then there exists a probability $p \in (0, 1)$ such that $B \sim pA + (1 - p)C$. That is, the DM is indifferent between B for sure and a suitable lottery between the best and worst outcomes.
- **Autor (2016) example:**
 - If a *bowl of miso soup* (A) is preferred to a cup of coffee (B), then a lottery giving *miso soup* with probability $1 - \varepsilon$ and *death by sushi knife* (C) with probability $\varepsilon > 0$ is still preferred to coffee, for sufficiently small ε .
- Continuity rules out **lexicographic preferences**
 - e.g. *Safety-first*: any positive probability of harm makes a lottery worse than safe alternative, regardless of other payoffs.
 - Violation of continuity, as $pA + (1 - p)C \sim B$ for **no** p .

Independence

- **Formal definition:** A preference relation $\succsim$ over lotteries satisfies **Independence** if for all lotteries $L, L', L'' \in \mathcal{L}$ and all $\alpha \in (0, 1)$,

$$L \succsim L' \iff \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''.$$

- Mixing both lotteries with the same third lottery L'' (with the same weight $1 - \alpha$) preserves their ranking.
- **Intuition:** Only the final outcome distribution matters — not the “structure” of the gamble. The DM does not care *how* uncertainty arises, only about the resulting probabilities.
- **Example:**
 - Suppose a DM prefers an **orange** (O) to an **apple** (A):
 $O \succ A$
 - Consider two new lotteries: Lottery 1: 50% Orange, 50% Pear; Lottery 2: 50% Apple, 50% Pear
 - By Independence, the DM must prefer Lottery 1 to Lottery 2, because both lotteries mix the original alternatives with the same “background risk” (Pear) with equal probability.

Example - Fancy Restaurant

- Assume a new restaurant opens in Halle with a **fancy concept**: Instead of choosing specific main courses, you can choose between random menus, e.g.
 - A** Starters, Main (50% probability tofu goulash with mushrooms, 50% vegetarian lasagna), Desert
 - B** Starters, Main (100% probability tofu goulash with mushrooms), Desert
- Using a vNM-utility function, you would calculate the utility of the options as
 - A** $u(A) = 0.5u(\text{Starters, Tofu, Desert}) + 0.5u(\text{Starters, Lasagna, Desert})$
 - B** $u(B) = u(\text{Starters, Tofu, Desert})$
- You should choose the option with **higher expected utility**.

Example - Fancy Restaurant (continued)

- Assume your decision is based on **heuristic** "*Choose the menu in which probability of Lasagna is as close to 50% as possible.*"
- This **decision wouldn't be in line with a vNM-utility function**. Why?
 - Assume the restaurant gets even fancier and mixes options A and B with option C (Starters, 100% Lasagna, Desert) at probabilities of 50%.
 - With the simple lottery, the heuristic would choose $A \succsim B$ (as in A probability of Lasagna = 0.5, in B = 0).
 - With the new compound lottery
 - A: $0.5A + 0.5C$, you'd get Tofu with probability 25% and Lasagna with 75%
 - B: $0.5B + 0.5C$, you'd get Tofu with probability 50% and Lasagna with probability 50%
- When C is mixed in, you'd thus switch from A to B, which contradicts the independence axiom.

Expected utility property

- For preferences that satisfy vNM EU theory, utility functions are **linear in the probabilities**:

$$U(\beta L + (1 - \beta)L') = \beta U(L) + (1 - \beta)U(L')$$

- **Important:** The vNM-utility function does not need to be linear in the bundles of goods, i.e., we do not need to impose $U(2L) = 2U(L)$ (*this depends on risk preferences*)
- **Risk preferences are captured by the shape of the utility function $U()$ (see Lecture 9)**

When can we apply expected utility theory?

- We need two ingredients to use expected utility theory:
 - ① a **utility function** that is defined over the certain outcomes of reduced lotteries, i.e. over bundles of goods. It captures preferences for risk and over bundles of goods.
 - ② the **vNM-assumptions on preferences over reduced lotteries** (rational and continuous preferences that fulfill the independence axiom).
- There are many situations in which these assumptions do not hold (*discussed in detail in Lecture 12*).
- If the necessary assumptions hold, this theory suffices to model **decisions under risk**.

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- Decisions under uncertainty: probabilities unknown $\rightarrow$ decision rules such as Maximin, Maximax, Minimax Regret.
- Decisions under risk: probabilities known or subjectively assessed $\rightarrow$ use expected values and expected utility.
- Lotteries: simple and compound lotteries; consequentialism ensures only reduced lotteries matter.
- Bayesian updating: new information updates beliefs via Bayes' Rule; strength of updating depends on baseline probabilities.
- Value of Information: information has value only if it changes optimal actions; perfect information provides an upper bound.
- Expected Utility Theory (vNM): under continuity and independence axioms, choices under risk can be represented by expected utility.

Important Keywords

- Uncertainty – unknown probabilities
- Risk – known (objective or subjective) probabilities
- Lotteries – simple vs. compound; reduced lottery
- Expected Value (EV)
- Bayes' Rule
- Value of Information
- Expected Utility (EU)
- vNM Axioms – continuity and independence

Questions?

